Optimal Racing Line, Energy Management, and Race Strategy for Electric Vehicles under Uncertain Grip

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Abstract

Autonomous electric racing demands joint optimization of trajectory, energy management, and race strategy — challenges compounded by uncertain tire grip and the nonlinear coupling between battery mass and energy consumption. We present a five-stage hierarchical framework: (1) time-optimal racing line computation via a custom Sequential Convex Programming (SCP) algorithm with a hand-coded Simplex solver; (2) Pareto-front analysis of the lap-time/energy tradeoff; (3) sampled worst-case robustification under uncertain grip; (4) battery sizing for multi-lap races under mass-energy coupling; and (5) race strategy co-optimization with a closed-form KKT solution for line choice and pace management. On a Monza-style circuit, SCP converges in 15 iterations to a 35.9 s lap time — substantially faster than SLSQP's ~180 iterations — while reaching a qualitatively similar solution. Grip uncertainty imposes a ~3 s lap-time penalty at the time-optimal point on the Monaco circuit, a cost that diminishes as energy efficiency is prioritized. Battery sizing yields a minimum feasible capacity of 37 kWh (824 kg) for a 51-lap race at full effort; pace management reduces this to 25 kWh, saving 60 kg of battery mass through analytically derived pace modulation under symmetric-race assumptions.

1. Introduction

Finding the fastest path around a race track is a classical problem in vehicle dynamics and motorsport engineering [1, 2]. The trajectory followed by a vehicle directly determines the curvature encountered along the track and therefore the maximum speed that can be maintained through each corner. By exploiting the full width of the track, drivers reduce curvature and increase cornering speed, leading to significant reductions in lap time.

For electric vehicles, racing line optimization becomes a multi-objective problem. In addition to minimizing lap time, the vehicle must manage a finite energy budget. Aggressive acceleration and high cornering speeds improve performance but increase energy consumption, while conservative driving strategies preserve battery charge at the cost of slower lap times. This tradeoff is particularly relevant in modern electric racing series such as Formula E [3] and in autonomous racing applications.

This work addresses the central question: *how should an electric autonomous race vehicle jointly optimize its trajectory, energy management, robustness to grip uncertainty, and race strategy?* We formulate a hierarchical optimization framework that progressively extends from minimum-time single-lap control to full race management. We study the tradeoff between lap time and energy consumption and analyze the impact of uncertain tire grip through a robust optimization framework, extending the single-lap analysis to full race management through battery sizing and strategy co-optimization.

The framework spans five stages: trajectory design via SCP with a custom Simplex solver, multi-objective Pareto analysis, sampled worst-case robustification under uncertain grip, race-level battery sizing with mass-energy coupling, and closed-form KKT race strategy. Stages 1–3 use a fixed 1600 kg kinematic model; Stages 4–5 adopt a 640 kg chassis with a mass-coupled battery model — a deliberate split discussed in the limitations. The optimized trajectories are subsequently tracked using a dynamic bicycle model and a Time-Varying Linear Quadratic Regulator (TV-LQR) to evaluate feasibility in closed-loop simulation.

2. Problem Formulation

2.1. Optimization Overview

The objective is to jointly optimize the geometric racing line and the corresponding speed profile around a closed circuit. We first compute the minimum-time trajectory considering only kinematic constraints, then extend the formulation to include battery energy consumption and generate a Pareto front, and finally account for uncertainty in tire grip. Figure 1 illustrates the track parameterization and the optimization variables.

2.2. Decision Variables

The racing line is parameterized using lateral offsets $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)$ measured from the track centerline. At station i, the vehicle position is $\mathbf{p}_i = \mathbf{c}_i + \alpha_i \hat{\mathbf{n}}_i$, where $\mathbf{c}_i$ denotes the centerline position and $\hat{\mathbf{n}}_i$ the outward normal direction. The constraint $|\alpha_i| \leq w_i$ keeps the path within the track of half-width w_i. The speed profile is $\mathbf{v} = (v_1, \ldots, v_n)$, and the complete optimization vector is $\mathbf{x} = (\boldsymbol{\alpha}, \mathbf{v}) \in \mathbb{R}^{2n}$.

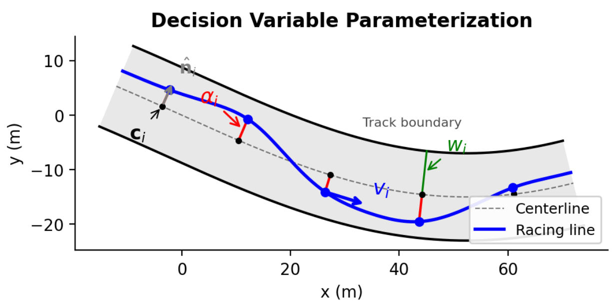


Figure 1: Track parameterization and optimization variables. The lateral offset α_i displaces each path point from the centerline $\mathbf{c}_i$ along the normal $\hat{\mathbf{n}}_i$. The speed v_i is along the tangent; w_i bounds the feasible region.

2.3. Vehicle and Track Dynamics

The optimization uses a kinematic point-mass model that captures the dominant physical limitations. At each station, the centripetal acceleration required to follow the curved path must not exceed the available tire friction:

$$v_i^2 \, |\kappa_i(\boldsymbol{\alpha})| \leq 0.85 \, \mu \, g \tag{1}$$

where 0.85 is a safety margin. Between consecutive stations, longitudinal dynamics are bounded by the vehicle's acceleration and braking capabilities:

$$v_{i+1}^2 \leq v_i^2 + 2 \, a_{\text{lon}} \, \Delta s_i, \quad v_i^2 \leq v_{i+1}^2 + 2 \, a_{\text{brk}} \, \Delta s_i \tag{2}$$

where $\Delta s_i = \|\mathbf{p}_{i+1} - \mathbf{p}_i\|$ is the segment arc length.

2.4. Lap Time and Energy Objectives

The total lap time is

$$T(\boldsymbol{\alpha}, \mathbf{v}) = \sum_{i=1}^{n} \frac{\Delta s_i(\boldsymbol{\alpha})}{v_i} \tag{3}$$

This objective is nonlinear in both $\boldsymbol{\alpha}$ (through Δs_i) and $\mathbf{v}$. At each segment, the traction force accounts for inertial, aerodynamic, and rolling resistance contributions: $F_i = m \, a_i + \frac{1}{2} \rho C_d A \, v_i^2 + C_r m g$. The mechanical power is $P_i = F_i \cdot v_i$. Depending on the sign of P_i, the electrical energy integrates P_i/η during drive phases or recovers $|P_i| \cdot \eta_{\text{regen}}$ during braking. Summing over all segments yields the total energy expenditure $E(\boldsymbol{\alpha}, \mathbf{v})$. All notation and vehicle parameters are gathered in Appendix A.

3. Time-Optimal Racing Line Using SCP

The first stage finds the time-optimal racing line and speed profile under constraints (1)–(2). Since both the objective and constraints are nonlinear, we use Sequential Convex Programming (SCP) following the trust-region approach in [4]. At each iteration, we linearize all nonlinear constraints around the current iterate and solve the resulting linear program (LP):

$$\min_{\Delta\mathbf{x}} \ \mathbf{c}^\top \Delta\mathbf{x} \quad \text{s.t.} \quad \mathbf{A}\,\Delta\mathbf{x} \le \mathbf{b}, \quad \boldsymbol{\ell} \le \Delta\mathbf{x} \le \mathbf{u} \tag{4}$$

where $\mathbf{c}$ is the linearized objective gradient. The constraint matrix $\mathbf{A} \in \mathbb{R}^{3n\times 2n}$ stacks the Jacobians of the cornering, acceleration, and braking constraints, and $\mathbf{b}$ contains the corresponding right-hand sides at the current iterate (see Appendix C for full expressions). The bounds $(\boldsymbol{\ell}, \mathbf{u})$ enforce both the trust region $|\Delta\alpha_i| \le \rho$ and the variable limits. We solve this LP with a two-phase Simplex algorithm implemented from scratch following [4]; a custom solver gives direct control over the trust-region stopping criterion and avoids the overhead of general-purpose LP interfaces. The curvature Jacobian $\partial\kappa_i/\partial\alpha_j$ is computed with JAX automatic differentiation. With $n = 80$ stations, the LP sub-problem involves $2n = 160$ decision variables and $3n$ linear constraints, keeping each iteration computationally lightweight compared to a full nonlinear solve.

The trust region radius ρ adapts at each iteration: halved when the step causes large constraint violations, grown by a factor of 1.2 when the step improves the objective. To validate the SCP solution, we solve the same problem with scipy's SLSQP. On the Monza-style circuit (80 stations, 1115 m), SCP converges from a curvature-minimizing warm start at 44.5 s to 35.9 s in 15 iterations, while SLSQP requires approximately 180 iterations to reach 35.4 s, consistent with both solvers converging toward a qualitatively similar optimum.

4. Time-Energy Pareto Optimization

The second stage addresses the tradeoff between lap time and energy consumption. For an electric vehicle that must complete multiple laps, aggressive time-optimal trajectories can discharge the battery rapidly and may not represent the best overall race strategy. To explore this tradeoff, we use the weighted-sum method [4]:

$$\min_{\boldsymbol{\alpha}, \mathbf{v}} \ (1-w)\frac{T}{T_{\text{ref}}} + w\frac{E}{E_{\text{ref}}} \tag{5}$$

where T_{ref} and E_{ref} are reference values from the time-optimal solution. Sweeping $w \in [0, 0.95]$ traces the Pareto front.

When we attempted to compute the Pareto front with SCP, the algorithm failed for most $w > 0$. As the energy weight increases, the optimal speed profile shifts far from the warm start. The linearized constraints become poor approximations, causing violations that trigger trust region shrinkage until convergence stalls. We therefore use scipy's SLSQP for the Pareto computation. The resulting Pareto front is shown in Figure 2b.

5. Robust Optimization under Uncertain Grip

In practice, tire grip varies along the track due to wet patches, rubber deposits, or temperature gradients. We model this by treating the friction coefficient as a per-station random variable: $\mu_i \sim \mathcal{N}(1.0, \sigma_\mu^2)$, clipped to

$[0.3, 1.5]$, with $\sigma_\mu = 0.1$.

We handle the stochastic constraints with a sampled worst-case robustification [4]: we draw $N = 20$ grip scenarios $\boldsymbol{\mu}^{(k)}$ and, exploiting the monotonicity of the cornering constraint in μ, tighten each constraint to the sampled worst-case grip at each station:

$$v_i^2\,|\kappa_i| \leq \mu_i^{\text{worst}} \cdot g \cdot 0.85, \quad \mu_i^{\text{worst}} = \min_k \mu_i^{(k)} \tag{6}$$

This reduces $N \times n$ scenario constraints to n deterministic constraints with modified grip, making the robust problem no more expensive than the nominal one.

6. Battery Sizing for Multi-Lap Racing

For a full race, the battery must store enough energy to complete all laps above $\text{SoC}_{\text{min}} = 5\%$. Battery capacity affects vehicle mass through $m = m_{\text{chassis}} + Q_{\text{batt}}/e_{\text{spec}}$, which in turn raises energy consumption per lap — making the sizing problem non-trivial. We sweep $Q_{\text{batt}} \in [15, 80]$ kWh; for each value, a forward–backward speed profile computes E_{lap} using a parabolic motor efficiency map (peak $\eta = 0.95$ at 60% rated power). A configuration is feasible if

$$Q_{\text{batt}}\,(1 - \text{SoC}_{\text{min}}) \;\geq\; n_{\text{laps}} \cdot E_{\text{lap}}(Q_{\text{batt}}) \tag{7}$$

and $Q^* = \min\{Q_{\text{batt}} : \text{feasible}\}$ is the optimal capacity.

7. Race Strategy Co-Optimization

We introduce two per-lap decision variables: a Pareto operating point $g \in [g_{\text{min}}, g_{\text{max}}]$ controlling cornering aggressiveness, and a pace factor $p \in [p_{\text{min}}, 1]$ scaling speed uniformly. Lap time and energy then scale as $T_{\text{lap}} = T_{\text{base}}(g)/p$ and $E_{\text{lap}} = E_{\text{base}}(g) \cdot p$, subject to the race energy budget:

$$\sum_{t=1}^{n} E_{\text{base}}(g_t)\,p_t \;\leq\; Q_{\text{batt}}(1 - \text{SoC}_{\text{min}}) \tag{8}$$

For a symmetric race, KKT stationarity and complementary slackness [4] yield the closed-form optimum: the energy constraint is active, and the optimal line minimizes $T_{\text{base}}(g) \cdot E_{\text{base}}(g)$:

$$g^* = \operatorname*{argmin}_g \; T_{\text{base}}(g)\,E_{\text{base}}(g), \qquad p^* = \frac{Q_{\text{batt}}(1 - \text{SoC}_{\text{min}})}{n \cdot E_{\text{base}}(g^*)} \tag{9}$$

with p^* clipped to $[p_{\text{min}}, 1]$ and a competitive constraint $T_{\text{base}}(g) \leq 1.15\,T_{\text{min}}$. An NLP validation (SLSQP) yields a locally optimal solution consistent with the constant (g^*, p^*) KKT prediction, supporting the closed-form result under the symmetric-race assumptions. Three assumptions underpin this result: (i) all laps are identical (symmetric race); (ii) the scaling $T_{\text{lap}} = T_{\text{base}}/p$ and $E_{\text{lap}} = E_{\text{base}} \cdot p$ is exact, which holds for uniform speed scaling but is an approximation when aerodynamic drag dominates; and (iii) $T_{\text{base}}(g) \cdot E_{\text{base}}(g)$ is unimodal in g over the feasible range, which is observed numerically but not analytically guaranteed.

8. Dynamic Trajectory Tracking

The kinematic optimization produces a racing line and speed profile, but these are not directly executable by a vehicle with tire slip and yaw dynamics. To bridge this gap, we formulate the trajectory tracking as a finite-horizon optimal control problem (full dynamics in Appendix B).

Given a reference trajectory $(\bar{\mathbf{s}}_k, \bar{\mathbf{u}}_k)_{k=0}^{N}$, we seek the control sequence that minimizes tracking error:

$$\min_{\mathbf{u}_{0:N-1}} \sum_{k=0}^{N-1} \left[(\mathbf{s}_k - \bar{\mathbf{s}}_k)^\top Q \, (\mathbf{s}_k - \bar{\mathbf{s}}_k) + (\mathbf{u}_k - \bar{\mathbf{u}}_k)^\top R \, (\mathbf{u}_k - \bar{\mathbf{u}}_k) \right] \tag{10}$$

subject to the bicycle dynamics $\mathbf{s}_{k+1} = f(\mathbf{s}_k, \mathbf{u}_k)$, where $\mathbf{s} = [x, y, \psi, v_x, v_y, \omega]^\top$ and $\mathbf{u} = [\delta, F_{\text{drive}}]^\top$.

We solve this in two phases. First, a Stanley lateral controller tracks the kinematic racing line to generate a dynamically feasible reference $(\bar{\mathbf{s}}_k, \bar{\mathbf{u}}_k)$ consistent with the bicycle model. Second, we linearize the dynamics around each reference point via finite differences to obtain $A_k = \partial f / \partial \mathbf{s}$ and $B_k = \partial f / \partial \mathbf{u}$, then solve the discrete Riccati equation backward to compute the TV-LQR gains K_k (see Appendix D). Online, the control law is

$$\mathbf{u}_k = \bar{\mathbf{u}}_k + K_k(\mathbf{s}_k - \bar{\mathbf{s}}_k) \tag{11}$$

with spatial indexing to handle timing mismatches between the reference and the actual vehicle state.

9. Results

9.1. SCP vs SLSQP Convergence

Figure 2a shows the convergence of both solvers on the Monza-style circuit (80 stations, 1115 m). Starting from a curvature-minimizing warm start at 44.5 s, SCP reaches 35.9 s in 15 iterations while SLSQP requires approximately 180 iterations to reach 35.4 s. The ~0.5 s gap (≈1.4%) reflects a quality-speed tradeoff inherent to SCP: the tight trust region prevents the LP sub-problem from over-shooting and enables early convergence, leaving a small residual suboptimality relative to SLSQP's tighter line-search, in exchange for a 12× reduction in iterations. A direct geometric overlay of the two racing lines would provide further validation, which we leave for future work.

9.2. Time-Energy Pareto Front

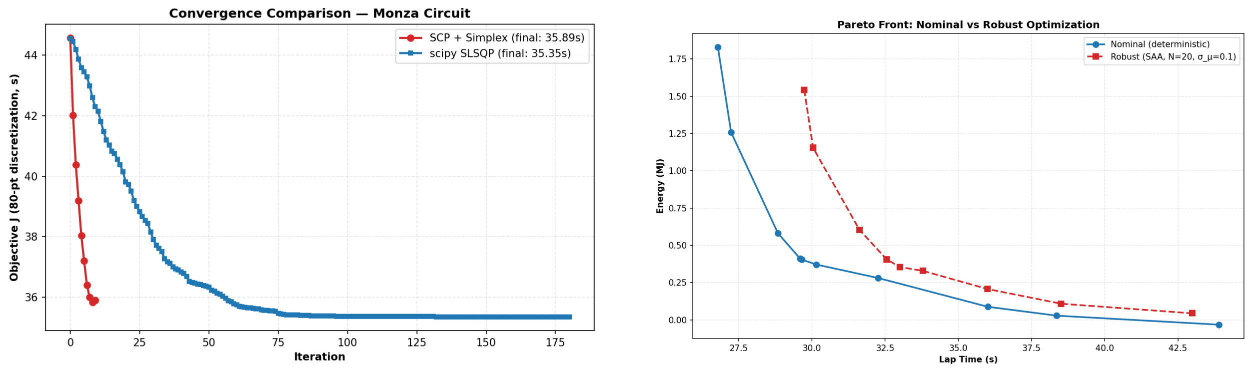
Figure 2b presents the nominal and robust Pareto fronts on the Monaco circuit. At $w = 0$, the nominal optimum reaches 27 s (consistent with an independent SCP run on this circuit). As w increases, the optimizer trades lap time for energy savings, reaching ~44 s at $w = 0.95$ with near-zero net energy. The robust front is shifted to longer lap times: at $w = 0$, the robust optimum is ~30 s versus 27 s, a 3 s penalty that quantifies the cost of robustness. However, this penalty shrinks as energy becomes more important in the objective, since energy-efficient trajectories are inherently smoother with lower cornering loads, reducing the sensitivity to grip uncertainty.

9.3. Nominal vs Robust Trajectories

Figure 3 compares the nominal and robust racing lines and speed profiles. The geometric racing lines are nearly identical since curvature reduction is grip-independent. The robust speed profile, however, is consistently lower, reflecting the conservatism of the sampled worst-case grip constraint. Yet the profile retains the same shape and variations as the nominal one: the vehicle still accelerates on straights and brakes into corners, only at reduced magnitudes.

9.4. Battery Sizing

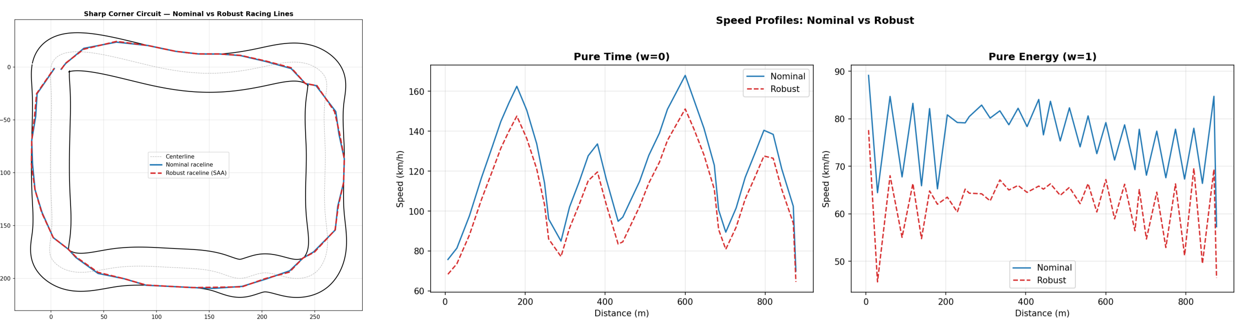
Figure 4 shows the battery sizing sweep for the Grand Prix circuit (1288 m, 51 laps, ≈65 km total). The left panel shows the energy balance as a function of Q_{batt}: below $Q^* \approx 37$ kWh, required energy exceeds available



(a) Convergence on the Monza-style circuit (80 stations, 1115 m).

(b) Nominal and robust Pareto fronts (Monaco circuit).

Figure 2: (a) SCP converges substantially faster than SLSQP while reaching a qualitatively similar solution. (b) The robust Pareto front is shifted to longer lap times due to worst-case grip constraints; the gap narrows for energy-efficient operating points.



(a) Racing lines on the track.

(b) Speed profiles along the track.

Figure 3: Nominal vs robust trajectories. The geometric paths are similar, but the robust profile maintains lower speeds throughout.

capacity and the race cannot be completed; above Q^*, a growing safety margin appears. At $Q^* = 37$ kWh the car mass is 824 kg and the final state of charge is 7.4%, just above the 5% threshold. The right panel confirms feasibility transitions sharply near Q^*: infeasible configurations finish with SoC $\leq 0\%$ while feasible ones grow linearly. The energy required per lap grows weakly with Q_{batt} because the mass-dependent energy increase is partially offset by richer regeneration at higher masses. The sharp transition near Q^* reflects the nonlinear feasibility boundary arising from this mass-energy coupling: a slightly undersized pack fails the race, while a slightly oversized one carries unnecessary dead weight.

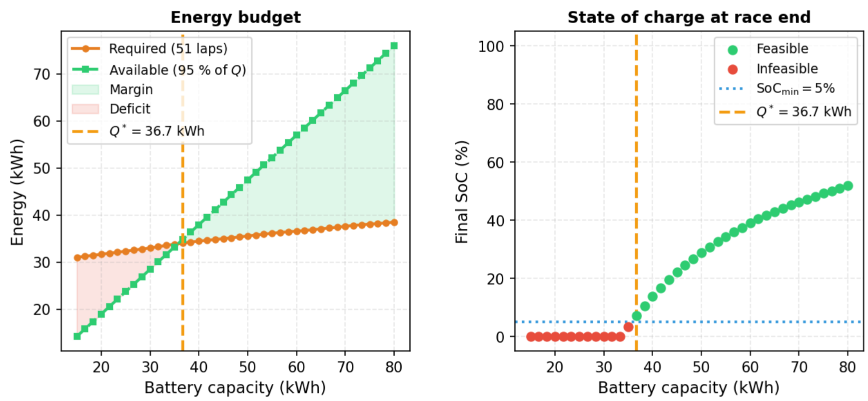


Figure 4: Battery sizing sweep for the Grand Prix circuit (51 laps, ≈65 km). Left: required vs. available energy as a function of battery capacity; $Q^* = 37$ kWh is the minimum feasible capacity. Right: final state of charge at race end; the sharp transition from 0% to positive SoC locates Q^* precisely.

9.5. Race Strategy, Battery Reduction, and Circuit Comparison

Figure 5 summarizes the race strategy co-optimization and circuit comparison. The left panel shows the time–energy Pareto front obtained by sweeping the grip fraction $g \in [0.50, 0.90]$: the fastest line ($g = 0.90$, $T = 50.2$ s) consumes the most energy (677 Wh/lap), while the slowest line within the competitive constraint ($T_{\max} = +15\%$) reduces energy by ~8%.

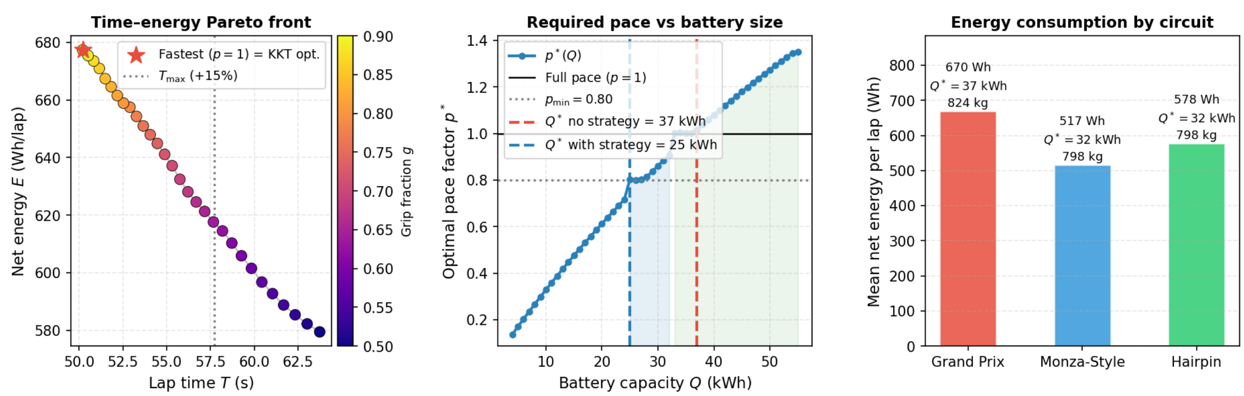


Figure 5: Race strategy and circuit comparison. Left: time–energy Pareto front; the fastest line within the competitive constraint ($T_{\max} = +15\%$) minimizes battery usage at full pace. Middle: optimal pace factor $p^*(Q)$; pace management reduces Q^* from 37 to 25 kWh (saving 60 kg). Right: mean energy per lap by circuit; the Monza-style circuit is 23% more efficient than the Grand Prix circuit due to lower curvature.

The middle panel shows $p^*(Q)$, the analytically optimal pace factor as a function of battery capacity. At $Q^*_{\text{baseline}} = 37$ kWh, full pace ($p = 1$) is required. With pace management ($p_{\min} = 0.80$), the minimum feasible battery shrinks to $Q^*_{\text{strategy}} = 25$ kWh, a 32% reduction that saves 60 kg of battery mass. The *strategy zone* (shaded blue) marks the range of Q where pace modulation is both necessary (car is too light to push full effort) and feasible (pace factor is above $p_{\min}$). This reflects how pace modulation decouples the energy budget from the timing constraint, enabling a lighter vehicle to remain competitive. The NLP validation (SLSQP over $2n$ per-lap decision variables) yields a locally optimal solution consistent with the constant (g^*, p^*) prediction (9), supporting the KKT analysis under the symmetric-race assumptions.

The right panel compares mean energy per lap across three circuits at ≈65 km race distance. The Grand Prix circuit (1288 m, 51 laps) is the most energy-intensive (670 Wh/lap), requiring $Q^* = 37$ kWh (824 kg). The Monza-style circuit (1115 m, 58 laps) consumes only 517 Wh/lap and requires $Q^* = 32$ kWh (798 kg), 23% more efficient due to lower curvature. The Hairpin circuit (1269 m, 51 laps) falls between at 578 Wh/lap and $Q^* = 32$ kWh. These differences underscore that circuit geometry — specifically corner frequency and severity — is the primary driver of battery sizing requirements.

10. Conclusion

This work presented a five-stage hierarchical optimization framework for electric vehicle racing. The SCP formulation efficiently solved the minimum-time problem (15 iterations vs. ∼180 for SLSQP on the Monza-style circuit), and multi-objective optimization revealed the time–energy tradeoff: small increases in lap time yield large energy savings. The robust sampled worst-case formulation added a 3 s penalty at the time-optimal point on the Monaco circuit, with the gap narrowing for energy-efficient operating points.

Battery sizing shows that a 37 kWh pack (824 kg) is needed to complete a 51-lap Grand Prix race at full effort. Race strategy co-optimization then reduces this requirement to 25 kWh (a 32% reduction, saving 60 kg) by exploiting the closed-form KKT solution to jointly optimize line choice and pace factor. Across three circuits at ≈65 km race distance, optimal battery capacity ranges from 32 to 37 kWh, driven by circuit curvature and its effect on energy consumption.

A key limitation is the gap between kinematic optimization and dynamic feasibility: the kinematic model overpredicts cornering speed because it ignores tire slip and yaw inertia. A more substantive limitation is the use of different vehicle models across stages: Stages 1–3 use a fixed 1600 kg kinematic model, while Stages 4–5 use a 640 kg chassis with a mass-coupled battery. The energy and sizing figures reported in Stages 4–5 therefore correspond to a different vehicle than the trajectory optimized in Stages 1–3, so the results are not fully self-consistent. A unified formulation would require iterating between trajectory optimization and battery sizing. Future work could unify these models, investigate chance-constrained formulations that reduce conservatism, relax the symmetric-race and uniform-pace-scaling assumptions in the KKT strategy (e.g., via dynamic programming over heterogeneous laps), and extend the race strategy to account for tire degradation or safety-car periods.

Contributions

Erwin Poussi developed the simulator, the SCP algorithm with Simplex and the scipy SLSQP pipeline, the iLQR trajectory tracking controller, the race simulation engine, and the stochastic optimization formulation. Matthieu Hautsch implemented the time–energy Pareto front analysis, the energy consumption model, the battery sizing sweep (Stages 4), and the race strategy co-optimization including the KKT analytical solution, the NLP validation, and the multi-circuit comparison (Stage 5). Code available at `https://github.com/Rwin2/EV-optimal-line-racing`.

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Supplementary Material

A. Notation and Vehicle Parameters

Symbol	Value	Description
m	1600 kg	Vehicle mass (Stages 1–3)
L	2.9 m	Wheelbase
l_f, l_r	1.4, 1.5 m	CG to front/rear axle
I_z	2500 kg m^2	Yaw moment of inertia
μ	1.0	Tire-road friction coefficient
C_f, C_r	80 000, 90 000 N/rad	Cornering stiffness (front/rear)
C_d	0.30	Aerodynamic drag coefficient
A	1.5 m^2	Frontal area
ρ	1.225 kg/m^3	Air density
c_r	0.012	Rolling resistance coefficient
P_{max}	250 kW	Maximum motor power
$F_{\text{drive,max}}$	8 000 N	Maximum traction force
$F_{\text{brake,max}}$	12 000 N	Maximum braking force
η	0.92	Motor efficiency
η_{regen}	0.65	Regenerative braking efficiency
g	9.81 m/s^2	Gravitational acceleration

Table 1: Vehicle parameters for Stages 1–3 (SCP, Pareto, robust optimization).

Symbol	Value	Description
m_{chassis}	640 kg	Chassis + motor + electronics mass
e_{spec}	200 Wh/kg	Battery system-level specific energy (Formula E Gen3)
P_{max}	300 kW	Maximum motor power
$F_{\text{drive,max}}$	10 000 N	Maximum traction force
$F_{\text{brake,max}}$	15 000 N	Maximum braking force
η_{peak}	0.95	Peak motor efficiency
η_{regen}	0.65	Regenerative braking efficiency
SoC_{min}	0.05	Minimum allowed state of charge
p_{min}	0.80	Minimum pace factor (strategy)

Table 2: Vehicle parameters for Stages 4–5 (battery sizing and race strategy), based on Formula E Gen3 regulations.

B. Bicycle Model Dynamics

The dynamic bicycle model used for trajectory tracking has state $\mathbf{s} = [x, y, \psi, v_x, v_y, \omega]^\top$ and control $\mathbf{u} = [\delta, F_{\text{drive}}]^\top$. The continuous-time equations of motion are:

$$
\begin{aligned}
\dot{x} &= v_x \cos\psi - v_y \sin\psi && (12)\\
\dot{y} &= v_x \sin\psi + v_y \cos\psi && (13)\\
\dot{\psi} &= \omega && (14)\\
\dot{v}_x &= \frac{1}{m}\left(F_{\text{drive}} - F_{\text{drag}} - F_{\text{roll}} + m\, v_y\, \omega - F_{yf} \sin\delta\right) && (15)\\
\dot{v}_y &= \frac{1}{m}\left(F_{yf} \cos\delta + F_{yr} - m\, v_x\, \omega\right) && (16)\\
\dot{\omega} &= \frac{1}{I_z}\left(l_f\, F_{yf} \cos\delta - l_r\, F_{yr}\right) && (17)
\end{aligned}
$$

where the lateral tire forces use a linear slip angle model with saturation:

$$
\begin{aligned}
\alpha_f &= \delta - \arctan\frac{v_y + l_f\,\omega}{v_x}, \quad \alpha_r = -\arctan\frac{v_y - l_r\,\omega}{v_x} && (18)\\
F_{yf} &= \text{clip}(C_f\,\alpha_f,\ -F_{\text{lim}},\ F_{\text{lim}}), \quad F_{yr} = \text{clip}(C_r\,\alpha_r,\ -F_{\text{lim}},\ F_{\text{lim}}) && (19)
\end{aligned}
$$

with $F_{\text{lim}} = \frac{1}{2}\mu\, m\, g$. The resistive forces are $F_{\text{drag}} = \frac{1}{2}\rho\, C_d\, A\, v_x^2$ and $F_{\text{roll}} = c_r\, m\, g$. The drive force is further limited by power: $F_{\text{drive}} \le P_{\max}/v_x$ when $F_{\text{drive}} > 0$. These equations are integrated using a fourth-order Runge-Kutta scheme at $\Delta t = 0.02\,\text{s}$.

C. SCP Sub-Problem Construction

At SCP iteration with current iterate $(\boldsymbol{\alpha}, \mathbf{v})$, the LP (4) is constructed as follows. Let $\Delta\mathbf{x} = (\Delta\boldsymbol{\alpha}, \Delta\mathbf{v})$.

Objective gradient. The linearized lap time gradient is:

$$
\mathbf{c} = \begin{bmatrix} \dfrac{\partial \mathbf{ds}}{\partial \boldsymbol{\alpha}}^\top \dfrac{1}{\mathbf{v}} \\ -\dfrac{\mathbf{ds}}{\mathbf{v}^2} \end{bmatrix} \tag{20}
$$

where $\mathbf{ds}$ is the vector of segment lengths and the Jacobian $\partial\mathbf{ds}/\partial\boldsymbol{\alpha}$ is computed from the path geometry.

Cornering constraints. Linearizing $v_i^2|\kappa_i| \le a_{\text{lat}}$ around the current iterate yields rows of $\mathbf{A}$ involving $v_i^2 \cdot \text{sgn}(\kappa_i) \cdot \partial\kappa_i/\partial\alpha_j$ (from JAX autodiff) and $2\, v_i\, |\kappa_i|$.

Acceleration/braking constraints. Similarly linearized from (2), coupling consecutive speed variables through the segment-length Jacobian.

D. TV-LQR Gain Computation

Given the reference trajectory $(\bar{\mathbf{s}}_k, \bar{\mathbf{u}}_k)$, we compute the discrete Jacobians $A_k \in \mathbb{R}^{6\times 6}$ and $B_k \in \mathbb{R}^{6\times 2}$ by finite differences of the RK4 integrator. The TV-LQR gains are obtained by solving the discrete Riccati equation backward:

$$
\begin{aligned}
K_k &= -(R + B_k^\top P_{k+1} B_k)^{-1} B_k^\top P_{k+1} A_k && (21)\\
P_k &= Q + A_k^\top P_{k+1}(A_k + B_k K_k) && (22)
\end{aligned}
$$

with terminal condition $P_N = Q$. The weight matrices are $Q = \mathrm{diag}(5, 5, 10, 1, 0.5, 0.5)$ and $R = \mathrm{diag}(10, 10^{-5})$, penalizing position and heading errors more heavily than lateral velocity, while keeping steering effort smooth.